

Knowledge graph approaches for linking datasets to uncover new biological insights

ECCB 2026 — Tutorial 12: Resources for plant sciences

Session 3 · 15:30–16:10

Dr. Carissa Bleker, National Institute of Biology, Slovenia

What this session covers

- Why knowledge graphs matter for your work
- What kinds of questions become answerable that weren't before
- A quick tour of the plant KG landscape
- **Hands-on:** querying real plant knowledge graphs yourself
- Where this is heading: natural-language access to graphs

The everyday problem

Data on a gene, pathway, or phenotype is scattered across:

- Pathway databases (KEGG, MetaCyc, AraCyc)
- Interaction resources (STRING)
- Published literature
- Your own omics/phenotyping data

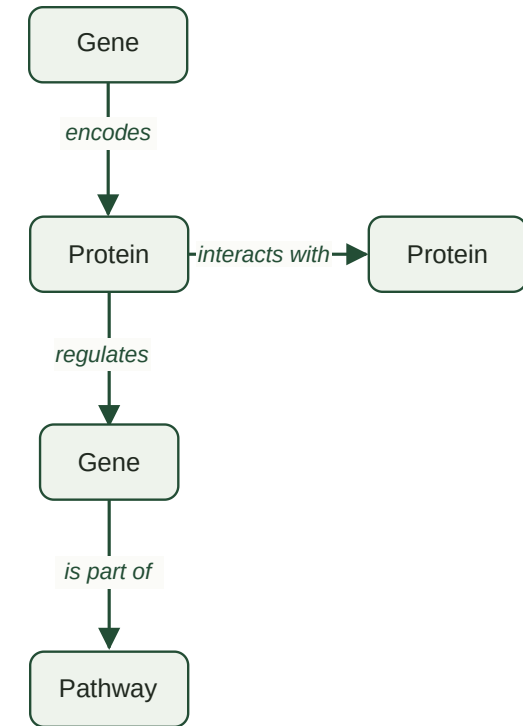
Each is searchable on its own, but **cross-cutting questions** integrating multiple sources of data are the hard part.



What is a knowledge graph?

A knowledge graph represents information as:

- **Nodes** — entities (a gene, a protein, a pathway, a phenotype, a paper)
- **Edges** — relationships between entities (*regulates*, *interacts with*, *is part of*)



Together: a traversable source of facts.

A concrete example

Much of the important information in biology lives in the *relationships*, not just the entities:

The **DREB2A** gene **encodes** a transcription factor that **regulates** drought-responsive genes

Represented as **triples**: Subject *predicate* Object

Gene *encodes* Transcription factor

Transcription factor *regulates* Target

ABA *is a* Phytohormone

Why this matters for research

Once relationships are explicit and typed, you can query *along* them.

Examples:

- Start at a stress phenotype → follow `associated with` → reach a candidate gene → follow `regulates` → get a list of **upstream regulators** to investigate
- Start at a significant locus (e.g. a GWAS hit) → follow `is part of` → reach the genes in that region → follow `interacts with` → narrow down to the most plausible **candidate genes**

These are single, easy-to-state graph queries, not a chain of separate database lookups stitched together manually

A step further: reasoning

Explicit relationships also let you **infer** new facts, not just retrieve stored ones:

- *Gene A* confers resistance to *Powdery mildew*
- *Powdery mildew* is a *Fungal disease*
- → Therefore: *Gene A* confers resistance to **some** Fungal disease(s)

This is what separates a knowledge graph from a static lookup table — connections you didn't explicitly store can still be derived from the ones you did.

Where this lives: graph databases

- Knowledge graphs are typically stored in a **graph database** (e.g. Neo4j) or as **RDF triples**, rather than rows and columns in a relational table
- Relationships are stored as **first-class objects**, so multi-step connections (gene → pathway → phenotype → condition) can be traversed directly
- Queried with graph-native languages — **Cypher** or **SPARQL** — though you don't always need to write these yourself



A Neo4j graph database showing labelled nodes (xxx , xxx) and typed edges (xxx , xxx)

What is an ontology?

- A controlled vocabulary for a domain: defines classes, subclass relationships, and how entities relate
- Matters in biology because the same thing can be named, grouped, or interpreted differently across datasets
- Example: "*drought tolerance*," "*water-deficit tolerance*," and "*drought resistance*" overlap in everyday speech, but aren't necessarily the same defined trait across phenotyping databases — an ontology (e.g. the Plant Trait Ontology) fixes what's meant and how it relates to other traits
- This is what lets a knowledge graph connect entities from different sources consistently

What a knowledge graph adds

A knowledge graph puts all of this in one place, connected, so you can ask questions that span sources:

- *What else is in the neighbourhood of my gene of interest?*
- *Is there existing evidence linking it to this stress response?*
- *Why might pathway A interfere with pathway B's effect?*

This is a shift from **lookup** to **traversal and reasoning**.



Question types this unlocks

- **Guilt-by-association** — find candidate genes/proteins via their network neighbourhood
- **Mechanism hypotheses** — find paths connecting an observed phenotype to known signalling components
- **Cross-omics contextualisation** — place your experimental hits (transcriptomics, proteomics) onto prior knowledge to see what's already known

These are the questions a single database or a literature search struggles to answer directly.

A few plant knowledge graphs to know

Resource	Built for	URL
AgroLD  AgroLD	Broad-scale Semantic Web integration across many crops	v2.agrold.org/agrold
KnetMiner  KnetMiner	Gene-centric candidate discovery, strong GWAS/QTL tie-in	knetminer.com
SKM 	Curated molecular interactions for plant stress signalling + hypothesis generation	skm.nib.si

AgroLD

- Semantic Web / SPARQL-based knowledge graph
- ~1 billion triples, 150+ datasets, 50+ plant species (broad crop coverage)
- Built to support hypothesis formulation and validation across diverse plant species
- Best fit when your question spans **many species or crops** and you're comfortable with SPARQL

v2.agrold.org/agrold



Placeholder: screenshot of the AgroLD homepage or Quick Search interface

KnetMiner

- Gene-centric network mining tool
- Evidence-scored search: ranks candidate genes by network support
- Strong integration with GWAS/QTL data
- Best fit when you're starting from a **trait or locus** and need **candidate gene prioritisation**

knetminer.com

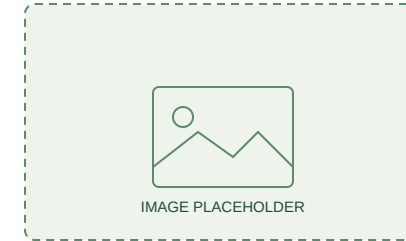


*Placeholder: screenshot of the KnetMiner
Plants Lite search/network view*

Stress Knowledge Map (SKM)

- Two complementary graphs:
 - **PSS** — curated, mechanistic *Plant Stress Signalling* model (800+ *reactions*)
 - **CKN** — Comprehensive molecular interaction *knowledge network* (26,000+ molecules, 480,000+ interactions)
- Built by domain experts through systematic literature/database curation
- Best fit for **plant stress biology**, mechanistic hypothesis generation, and quantitative modelling

skm.nib.si



Placeholder: screenshot of the SKM PSS or CKN web app

Let's try them

We've seen what each resource is built for — now let's actually use them.

Hands-on: AgroLD

Using the browser interface:

- **Quick Search** — keyword search across the whole knowledge base (e.g. a gene name)
- **Advanced Search** — pick an entity type (gene, protein, pathway...) and search within it, results as a sortable/downloadable table
- **SPARQL Query Editor** — comes with ready-made example queries you can run as-is, e.g. *"find all genes involved in the Calvin cycle"* — no need to write SPARQL yourself



v2.agrold.org/agrold

Hands-on: KnetMiner (Plants Lite)

Using the browser interface. Select the (free) **Plants Lite** resource:

- **Keyword search** — e.g. a trait of interest ("*grain colour*", "*drought tolerance*")
- **Gene list / genome region search** — start from your own candidate genes or a genomic interval
- **Network view** — launch the knowledge network for a gene, explore connections, and see the evidence-ranked candidate list

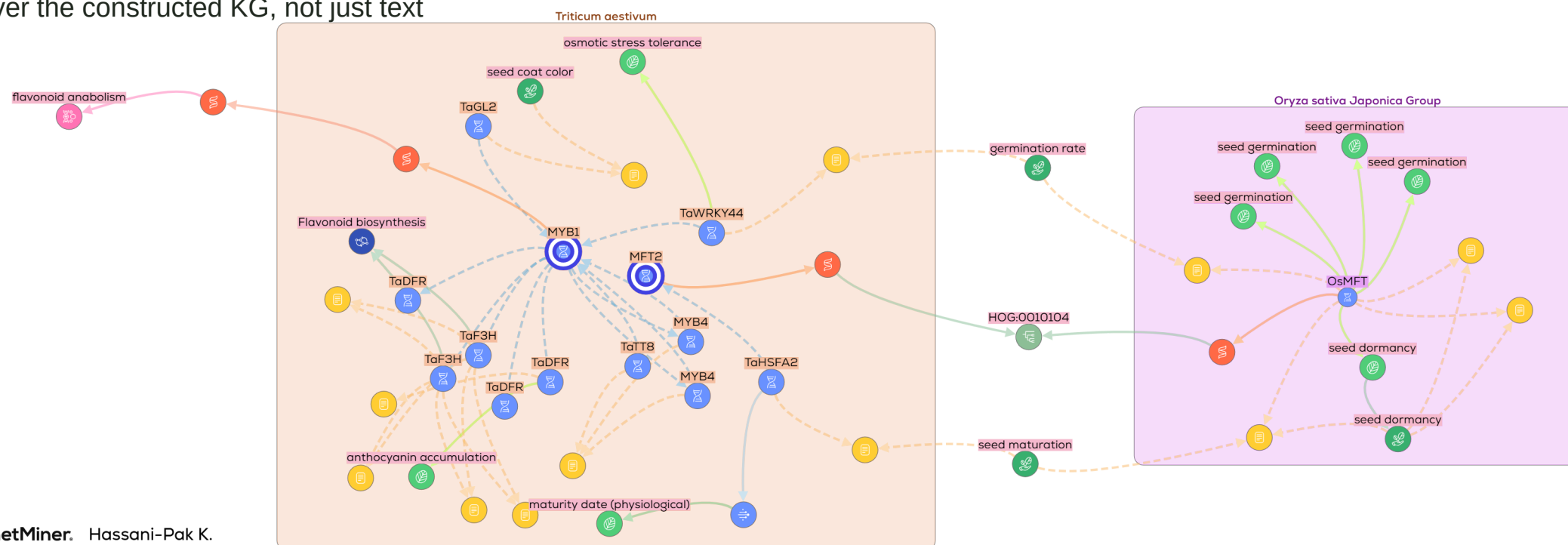


app.knetminer.com/plants-lite

Worked Example: Grain Colour & Pre-Harvest Sprouting (PHS)

Start from a **wheat gene list** and the traits "grain colour and pre-harvest sprouting":

- **Trait dissection** — KnetMiner breaks the trait into sub-components (seed dormancy, testa pigmentation, ABA signalling) rather than treating it as one keyword
- **Semantic motif search** — runs 180+ pre-defined motif queries per gene, scoring each for evidence linking it to grain colour, dormancy, and sprouting resistance
- **Interactive gene-trait network** — builds a knowledge graph connecting genes, QTLs, publications, and ontology terms, ranked by evidence
- **Graph Chat** — ask follow-up questions in natural language ("*which of these genes also affect root architecture?*"); the agent reasons directly over the constructed KG, not just text



Hands-on: SKM

Using the browser interface:

- Search for a gene/protein of interest and explore its neighbourhood interactively
- Visualise and export subnetworks directly from the browser

skm.nib.si/pss OR skm.nib.si/ckn

This covers most everyday questions. For more targeted, repeatable, or programmatic interrogation, we turn to **skm-tools** next.



Hands-on: querying SKM yourself

We'll now query SKM directly using [skm-tools](#), a Python toolkit built for the purpose of interrogating PSS and CKN beyond what the web app supports.

Notebook: `placeholder link to notebook on GitHub`

To run:

Option A — Google Colab (no install, runs in browser)

`placeholder link`

Option B — Local Python

```
pip install "skm-tools @ git+https://github.com/NIB-SI/skm-tools.git"
```

Note: Visualisation of network analyses results in Cytoscape requires a local Cytoscape install and won't work in Colab.

What we'll do in the notebook

1. Load a SKM graph as a NetworkX object
2. Filter to the interactions relevant to our biological question
3. Extract neighbourhoods around genes/components of interest
4. Find paths connecting two parts of the network
5. Visualise the resulting subnetwork



Case study — [placeholder]

The future? Beyond manual querying

What we just worked through required either a developer to have implemented the functionality in a user interface (search interface, API, ...), or *you* to write queries/scripts and manually chain results together for any multi-step question.



Knowledge graphs are a strong foundation for retrieval-augmented generation (RAG) — letting an LLM do that work instead:

- Selective retrieval of structured content from an external resource
- Answers stay grounded in explicit graph facts, not free-text guessing
- Natural language interaction with the graph's content, while maintaining the traceability of a real query — without anyone having to write one

The goal: the same trustworthy, structured answers, but without a human needing to manually preparing queries and analyses.

Where this fits in the bigger picture

- Knowledge graphs contextualise genes, pathways, and phenotypes within existing knowledge
- They're one route to making sense of large-scale data — including pangenome and variant data

Next up (Session 4): building and evaluating plant pangenomes — another way of structuring large-scale plant data for reuse.



Thank you

Resources mentioned

- SKM: skm.nib.si, skm-tools: github.com/NIB-SI/skm-tools
- AgroLD: v2.agrold.org/agrold
- KnetMiner: knetminer.com

Several slides on knowledge graph basics and ontologies adapted from BioCypher Workshop materials (ssciwr/slides-biocypher), CC-BY 4.0

Questions welcome

Contributors